

Measuring Manhood

Race and the Science of Masculinity,
1830–1934

Melissa N. Stein



University of Minnesota Press
Minneapolis
London

Copyright 2015 by the Regents of the University of Minnesota

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without the prior written permission of the publisher.

Published by the University of Minnesota Press
111 Third Avenue South, Suite 290
Minneapolis, MN 55401-2520
<http://www.upress.umn.edu>

Library of Congress Cataloging-in-Publication Data

Stein, Melissa N.

Measuring manhood : race and the science of masculinity, 1830-1934 / Melissa N. Stein.

Includes bibliographical references and index.

ISBN 978-0-8166-7302-5 (hc)—ISBN 978-0-8166-7303-2 (pb)

1. Racism—United States—History. 2. Masculinity—United States—History.
3. Sexism—United States—History. 4. Individual differences—Social aspects—
United States—History. 5. Individual differences—Political aspects—United States—
History. 6. Science—Social aspects—United States—History. 7. Sociobiology—United
States—History. 8. United States—Race relations—History. 9. United States—Social
conditions—1865-1918. 10. United States—Social conditions—1918-1932.

I. Title.

E184.A1S793 2015

305.800973—dc23

2014043033

Printed in the United States of America on acid-free paper

The University of Minnesota is an equal-opportunity educator and employer.

21 20 19 18 17 16 15 10 9 8 7 6 5 4 3 2 1

For my family—past, present, and future

Contents

Introduction: Making Race, Marking Difference	1
1. "Races of Men": Ethnology in Antebellum America	27
2. An "Equal Beard" for "Equal Voting": Gender and Citizenship in the Civil War, Reconstruction, and Redemption	89
3. Inverts, Perverts, and Primitives: Racial Thought and the American School of Sexology	169
4. Unsexing the Race: Lynching, Castration, and Racial Science	217
5. Walter White, Scientific Racism, and the NAACP Antilynching Campaign	251
Epilogue	277
Acknowledgments	285
Appendix. Charting Racial Science: Data and Methodology	289
Notes	293
Index	349

Making Race, Marking Difference

IN A 1994 SPEECH BEFORE the Rutgers Council of the American Association of University Professors discussing college admissions and diversity programs, Rutgers University president Francis L. Lawrence remarked that minority students lacked the “genetic hereditary background” to excel on standardized tests—a comment for which he later apologized, insisting that he misspoke.¹ Similarly, Harvard University president Larry Summers suggested at a 2005 conference on diversity in science careers that “socialization and continuing discrimination” were perhaps less responsible for the small representation of women in the field than women’s own biologically determined capacities (or incapacities). That is, women simply had less of an innate aptitude for certain scientific professions than men.² Summers’s comments were eerily reminiscent of another respected Harvard man more than a century prior: Dr. Edward Clarke, a professor in the university’s medical school and an opponent of higher education for women. In his tremendously influential 1873 publication, *Sex in Education, or, A Fair Chance for the Girls*, Clarke insisted that women’s bodies were not designed for intellectual pursuits, and as a result, higher education would impinge on their true biological function—reproduction—by diverting blood from their wombs to their brains, in turn jeopardizing not only their own health but also the future of the (white) race.³ Going back farther still in the hallowed halls of Harvard, we find Louis Agassiz, a revered Swiss-born professor of zoology and prominent racial theorist. Agassiz was famous in the mid-nineteenth century for his contributions to natural science and innovations in teaching. He was also known among his contemporaries for his marriage to Elizabeth Cabot Cary, famous in her own right as a writer and advocate of women’s education, who, following her husband’s death, led the charge to open Harvard’s doors to women. She ultimately succeeding in 1879—the same decade her colleague Edward Clarke published *Sex in Education*—and later became the first president of Radcliffe College, Harvard’s “sister” school. But Agassiz remains most well

known today for his vocal support for the theory of polygenesis, which claimed that the “races” originated separately and that each race’s unique biological makeup had not and would not change over time—a theory that in the antebellum period in particular often served as a scientific justification for America’s racial hierarchy.

This is intended not to disparage Harvard University (or Rutgers, my own graduate alma mater) but rather to show how *mainstream* biological determinism is in regards to both race and gender that one might hear the ethos of “biology is destiny” espoused at even the most esteemed institutions in the United States and also how *persistent* such ideas have been since the nineteenth century. In the case of both university presidents in the 1990s, the response to their comments—particularly from women’s and minority rights groups—was swift and very public. However, neither of the men lacked defenders, some of whom were quick to assert that the comments were well supported by science, seemingly an entity unto itself that cared little for liberal agendas and only concerned hard facts. Indeed, though more likely to be challenged today (particularly when articulated as baldly as Lawrence and Summers), the belief that race and gender, as well as sexuality, are biological categories never completely disappeared. In fact, such beliefs have seen a resurgence in the United States in the twenty-first century, albeit often in more implicit ways and with different cultural or political aims than the long nineteenth century—the ways in which studies of the “female brain,” for example, continue to be employed toward questions of educational policy and classroom practice.⁴

In regards to sexuality in particular, television and print news media regularly report on scientific studies seeking a biological basis of homosexuality. From the search for the “gay gene” to a widely reported story that linked homosexuality to the ratio of index to ring finger lengths, scientists have continued to scrutinize every part of the body for clues to sexual orientation.⁵ In “Born Gay? How Biology May Drive Orientation,” a lengthy article in the *Seattle Times* that described the finger length studies, among other scientific investigations on homosexuality and heredity, science reporter Sandi Doughton clearly articulated the sociopolitical context—and applicability—of such work: “The social and political implications of the research are impossible to ignore, leading to unease on both sides of the gay-rights debate. If science proves homosexuality is innate, is there any basis to deny gays equal treatment—including the right to marry? But if scientists unravel the roots of sexual orientation, will it some day be

possible to ‘fix’ people who don’t fit the norms or abort fetuses likely to be born gay?”⁶ While much has changed in American culture since the turn of the twentieth century, the scientific debate Doughton depicted in 2005 shared more than a passing resemblance to medical scientists’ considerations of the “congenital” or “acquired” nature of homosexuality nearly a hundred years earlier; for example, in an article in the *Medico-Legal Journal*, respected New York gynecologist and sexologist Bernard Talmey sought to determine whether one of his patients demonstrated “a case of inherited or acquired homosexuality” and found that indeed the man’s “entire clan was sexually tainted.”⁷ Likewise, the concern Doughton raises about biologically deterministic sexual science today underscores that the “nature or nurture” question in regards to homosexuality also continues to have eugenic implications.⁸

The dawn of the twenty-first century has also seen a resurgence of scientific interest in biology and race. Biomedical researchers have devoted considerable energy to comparing disease rates across racial groups, and in 2005, the Food and Drug Administration approved the first race-specified drug, BiDiL, marketed by the pharmaceutical company Nitromed exclusively to African Americans for heart failure. In addition to ethical concerns and uncertainty about its effectiveness, scientific and public debates over the controversial drug have frequently raised the question “Is race biological?”⁹ Advances in DNA research have also precipitated scientific investigation into race and racial histories. In conjunction with the African American Roots Project, which provides similar services for members of the general public, Harvard professor Henry Louis Gates Jr. combined genealogical research and DNA mapping to trace the ancestry and family histories of famous African Americans in a series of television documentaries. In the nineteenth century, racial theorists used science to deny black people a history, any claim to civilization, and even a place in the human family. There is certainly a poignant irony, then, in science being used to help African Americans reclaim their past. To be sure, the cultural politics driving this work on DNA or on disease are quite different from the nineteenth- and early twentieth-century racial science described herein. Still, any slippage between race as a social category and biological entity in scientific endeavors, no matter how noble the intentions, raises important questions and concerns—not the least of which is that, as this book demonstrates, any conclusion implying racial difference of some kind could have a life well beyond the scientist’s intent.¹⁰

This book describes how race became the purview of science and the processes by which race was constructed as a biological phenomenon with far-reaching social, cultural, and political resonance. In so doing, it tells the story of an overlapping, interdisciplinary group of scientists who asserted their relevance and authority by offering expert advice on America's most pressing issues. These scientists often used gender and sex difference to buttress their claims about racial difference, but the mechanisms by which they did so constantly shifted according to what was at stake in that specific historical moment. But throughout the nineteenth and early twentieth centuries, their conclusions about human difference helped naturalize and institutionalize sociopolitical difference—and hierarchy—in America. For these scientists, the physical body both reflected and determined the character of the social body.

But why should we care about a hodgepodge group of long-dead scientists, whose work in some cases seems so absurd and blatantly biased by contemporary standards that it is all too easy to dismiss? For one, they are still making race—and gender—trouble today. Scientists who find evidence of biological differences between the races, or between the sexes, continue to make front-page headlines and spark debate among talking heads on television shows targeted to mainstream audiences, while the countless studies that challenge those findings or the implications drawn from them receive far less public attention. From the “gay gene” and the “female brain” to minority students’ insufficient “genetic hereditary background,” we are still dealing with the legacy of scientific ideas that coalesced in the nineteenth century. This project, then, seeks to unpack their roots, as well as their impact in U.S. history.

More specifically, I am concerned with not only ideas and discourses but also how the public received such ideas and how they shaped the way real people were treated. Accordingly, this book examines a variety of sources, both textual and visual, to interrogate racial scientists’ engagement with social and political issues as well as the incursion of scientific thought into political culture. I first survey the ways in which gender and sex were implicated in the rise of race as a subject of *scientific* inquiry and then focus on a number of key moments from the heyday of biological determinism—roughly concurrent with the long nineteenth century—that are particularly illustrative of the complex intersections of race, gender, and sexuality in medico-scientific thought; the sociopolitical

import of human bodies; and the ways in which scientists sought to establish and defend their authority over Americans’ most pressing concerns.

The centrality of race in public and scientific discourse reached an apex at the turn of the twentieth century, a time during which the body was imbued with enormous social meaning and significance. Within the multiple and intersecting discourses about bodies—evolution, hysteria, pathology, sexual difference, masculinity—race remained, to borrow Evelyn Brooks Higgenbotham’s useful terminology, the metalanguage that wove those discourses together.¹¹ Race determined how bodies would be interpreted and also how they would be treated as a result. Moreover, as historian Laura Briggs notes, “Race is not an attribute that inheres in bodies, but rather attaches itself to bodies through the ideological and material work of things like law, medicine, science, economy, education, literature, social science, public policy, and popular culture. Race in turn fundamentally shapes both the subject and content of those diverse fields.”¹² As we shall see, race provided the *raison d’être* for whole fields of scientific inquiry and a space for scientists to lay claim to social and political relevance.

However, race itself was still being *made* as a category of medico-scientific analysis during much of the nineteenth century, and early scientists of race relied on more familiar categories of human difference to conceptualize and describe it. *Gender*—a metalanguage in its own right and one already in use as a biological means through which the populace could be parsed and understood—functioned as a rhetorical tool for scientists to employ in constructing race. That is, while it is common for scholars to describe intersecting differences as mutually constitutive, when it comes to the role scientists and physicians played in making and naturalizing race, it would be more precise to say gender difference (which predated the shift during the early modern period in the meaning of the word “race,” from lineage and bloodlines to the more recent definition regarding skin color and physical features) offered scientists a convenient reference point and foundation through which to frame their ideas about *race* as a nascent category of difference similarly ordained by nature itself.¹³ At a time when most Americans found it difficult to imagine human beings without regard for gender or sex—categories to which many social and political rights were attached—racial scientists often maintained that blacks deviated from white standards of both gender performance and bodily sex to call into question their very humanity.

Race in Early America

During the nineteenth century, racist ideologues increasingly drew upon science for legitimacy and authority, whereby popular ideas about the roles and capacities of men and women were naturalized as biologically determined when science came of age in earlier centuries.¹⁴ Racism was not new, but as George M. Fredrickson points out in his seminal text *The Black Image in the White Mind*, the type of racism that gained strength throughout the nineteenth and into the early twentieth centuries was “a rationalized pseudoscientific theory positing the innate and permanent inferiority of nonwhites.” Fredrickson distinguishes this new scientific racism from the “protoracist” prejudice and discrimination of prior centuries—that is, what might be better interpreted as xenophobia.¹⁵ His use of the term “pseudoscience” is somewhat misleading, however. Largely forgotten today, ethnology, or the “science of race” as defined by its early champions, was familiar to many in the nineteenth century. In his influential 1854 tome *Types of Mankind*, Alabama physician Josiah Nott declared ethnology to be “eminently a science for American culture.” Indeed, proponents of ethnology generally read into the body ideas about race that already permeated the national imaginary. Racial science shaped and was shaped by American race relations and political rhetoric. Scientists frequently used gender and sex difference to bolster their claims about racial hierarchy and police the category of citizenship, but precisely how sex and race intersected in scientific thought changed significantly over time, corresponding to specific sociopolitical concerns among the scientists.

Though Nott was careful to define the term “ethnology” for his mid-nineteenth-century readers, the principles of scientific racism were by then at least a half-century old, even if ethnology as a defined field was still developing. In the early colonial period in North America, European colonists initially distinguished themselves from Africans and Native Americans not on the basis of color distinctions but rather through asserting religious and cultural differences. However, since cultures could adapt and people could convert, it soon became clear that society needed to be organized around differences that were both more visible and permanent if Europeans were to build and maintain their own power, authority, and dominance. Color filled the need for a means of conferring status that was immutable and readily apparent, and color thus gradually became the

foundation through which difference was constructed and maintained. Over time, Americans no longer considered themselves Anglo-Christians, distinguished from the heathens in their mix, but as “white.” An inchoate and amorphous category, whiteness was defined not so much by what it was as by what it was not—Native American or African. More important, though, in the developing nation leading up to and after independence, “white” was a category with increasingly tangible rewards, including the rights and privileges of citizenship.¹⁶

The birth of the United States represented a pivotal and complex moment in race making. The language of equality, protection from tyranny, and the “rights of man” used in American Revolutionary rhetoric seemed to conflict with the reality of a slave society—an incongruity that eighteenth-century African Americans certainly noted. However, many historians have argued that the Founding Fathers actually intended their vision of a democratic society to apply only to white, propertied men. Most notably, Thomas Jefferson, who wrote so eloquently about the ideals of democracy in the Declaration of Independence, was himself a slave owner and yet publicly expressed deep ambivalence over slavery. Despite conflicted sentiments over the institution, Jefferson proposed in his widely read text *Notes on the State of Virginia*, first published in the early 1780s, that blacks were most likely naturally and irreconcilably inferior. In his *Notes*, Jefferson, an early theorist of scientific racism, applied the principles of animal husbandry to a lengthy and strange discussion of racial mixture and expressed clear distaste for black physiognomy, which he deemed an “eternal monotony” and an “immovable veil of black which covers all the emotions of the other race.”¹⁷

A virtual microcosm of the racial tropes and themes that later characterized ethnology, Jefferson made a number of claims in *Notes* about black people specifically and about racial difference more generally that would continue to echo in scientific writing on race well into the twentieth century. Notably, gender played a key role in Jefferson’s assessment of racial difference and hierarchy. “A black, after hard labour through the day, will be induced by the slightest amusements to sit up till midnight, or later, though knowing he must be out with the first dawn of the morning. They are at least as brave [as white men], and more adventuresome. But this may perhaps proceed from a want of forethought,” he wrote. His argument here foreshadowed paternalistic defenses of slavery by antebellum scientists; black men were suited to hard labor but impetuous and incapable

of acting in their own best interests. Even areas in which they seemed to compare favorably with white men—"they are at least as brave, and more adventuresome"—instead underscored racial difference and gender deviance. Bravery in white men was simply that, while apparent bravery in black men stemmed from their recklessness, impugning their manhood rather than recommending it.

Most important among his many claims about race, though, was that racial differences were not cultural but rather natural and rooted in the body itself. This belief in particular drove racial science and remained a central tenet in the field even as other concerns, debates, and frameworks in the field changed over time. Though Jefferson expressed some uncertainty as to exactly from where in the body race sprung, he was an early proponent of the idea that it was both biological and pervasive. Jefferson also assumed racial difference held tremendous importance. "This unfortunate difference of colour, and perhaps of faculty, is a powerful obstacle to the emancipation of these people," he maintained. Moreover, he found the black and white races so different that he suggested that they might be entirely different species. Near the end of his discussion of slavery and race in *Notes*, Jefferson offers a call to arms for further study into racial difference, writing, "It is not against experience to suppose, that different species of the same genus, or varieties of the same species, may possess different qualifications. Will not a lover of natural history then, one who views the gradations in all the races of animals with the eye of philosophy, excuse an effort to keep those in the department of man as distinct as nature has formed them?"¹⁸ Scientists began to take up Jefferson's call shortly thereafter and more widely still in the nineteenth century.

Scientific Bodies

But it is also important to note that Jefferson himself was writing at a critical moment in the development of the biological and natural sciences. The eighteenth century was the Age of Enlightenment, the Age of Reason. Science gained a new place of prominence and respect in Europe and North America, with particular emphasis on understanding the natural world and, in the wake of Swedish naturalist Carl Linnaeus's taxonomy system, classifying it as well. As Londa Schiebinger demonstrates in *Nature's Body: Gender in the Making of Modern Science*, gender profoundly shaped how scientists viewed and understood that natural world. Plants took on

gendered human characteristics in botany texts that read like romantic novels, for example. Human beings, of course, were a part of the natural world, and scientists were also increasingly interested in biological differences between men and women and the implications of anatomy for their place in the social order. Thomas Laqueur's *Making Sex: Body and Gender from the Greeks to Freud* examines an eighteenth-century shift from a one-sex model of sexual difference in medical and cultural discourse, whereby women were viewed as imperfect, underdeveloped men rather than a separate sex, to a two-sex model with women and men positioned as distinct and opposing sexes. This shift illustrates Laqueur's assertion that the body, and seemingly immutable anatomy itself, is viewed through a cultural lens. In both models, anatomists and medical scientists interpreted the body and sex difference in such a way that they reinforced the existing cultural and political ideology about gender roles in society. In other words, the sexed body both reflected and determined the gendered sociopolitical order, regardless of the era in which it was conceived. For Laqueur, sex, like gender, has been—and continues to be—made.

Other scholars have similarly demonstrated the ways in which the scientific body is inextricably tied to the historical and cultural context in which scientists interpret it. In examining the changing medical treatment of intersexuality in England and France in the nineteenth and early twentieth centuries, historian Alice Dreger reveals how scientists and physicians have at times quite literally constructed sex. With advancements in medicine and surgical technology, physicians attempted to manipulate sexually ambiguous bodies to fit into increasingly rigid gender roles—though the people inhabiting those bodies were often more resistant than they expected.¹⁹ Echoing Dreger, as well as Schiebinger and Laqueur, Lisa Moore and Adele Clarke's "Clitoral Conventions and Transgressions: Graphic Representations in Anatomy Texts, c. 1900–1991" argues that anatomy, or specific parts of the body, can take on enormous cultural significance and reflect the gendered assumptions of both the scientists and the society in which they are writing.²⁰ Anne Fausto-Sterling's *Sexing the Body* unpacks various theories of sex difference from scientific, medical, philosophical, and historical texts and critiques what she sees as the false dichotomy between nature and nurture, asserting that the boundaries between the body and the world outside are more permeable than we think. Most notably, Sterling proposes that sex, like gender, be read as a spectrum or fluid continuum. The body is a site of knowledge production,

she maintains, and science is the tool with which it is produced—though never in isolation from cultural influences. Moreover, she notes that the more feminists have attacked the norms and limitations of gender, the more scientists have been determined to find and prove biological differences of sex in an attempt to naturalize social distinctions or inequalities. As I demonstrate, scientific studies of race in the long nineteenth century followed a similar trajectory. While scientists continuously drove American discourse on race in the nineteenth century, they were also in many regards reactionary. As nonwhite races—and women—resisted the sociopolitical limitations they faced and challenged white male dominance, scientists rallied to ground their power in biology.

Jefferson's *Notes on the State of Virginia* was very much a product of the eighteenth-century scientific genealogy that Laqueur and Schiebinger describe, but it also anticipated the direction science would take in the nineteenth century. With a well-established model of using the body to naturalize gender roles and justify social divisions based on sex, scientists began to utilize the logic of "biology is destiny" in regards to race. Echoing the notion of gender differences rooted in biology that could never be legislated away, scientists in Jefferson's wake moved increasingly away from environmental concepts of race and spoke instead of the immutable and "natural" corporeal differences of African Americans in particular, differences that irrevocably determined their social inferiority. As the male–female binary was increasingly bifurcated and "legitimized" by scientists, so too was the binary of black–white. Grappling with the kinds of questions Jefferson had raised, the field of ethnology—which its adherents defined as the "science of race"—developed gradually over the first decades of the nineteenth century and coalesced in the United States in the 1830s with the work of Samuel Morton and the elaborate racial taxonomy he created based on his enormous collection of human skulls, discussed in chapter 1.

Europe had long seen a booming scientific literature on race dating to the early modern period, prompted by the diverse new peoples Europeans continuously encountered during the age of exploration and colonization.²¹ Numerous scholars have pointed out that these explorers seemed particularly fascinated by Africans' sex organs and sexual practices.²² Europeans' scientific and popular fascination with sexual aspects of race later culminated in—and was exemplified perhaps most famously by—the tragic story of Saartjie Baartman, a young South African woman who was

brought to London in 1810 and publically exhibited across Western Europe as "the Hottentot Venus," a living specimen of sexualized exoticism and racial difference.²³ In the late eighteenth and early nineteenth centuries, European scientists also increasingly looked to the newly formed United States of America as a perfect laboratory for their theories about race, while at the same time, American science was coming into its own and a growing number of U.S. scientists added their voices to the transatlantic conversation, with Morton among the most publicized. Indeed, contemporaries and modern scholars alike have characterized Morton as the founding father of the American school of ethnology, which was largely synonymous with the theory of polygenesis (the separate origins of the races) but also characterized by its proponents applying biology and history to issues specific to the U.S. racial context, such as chattel slavery and political debates over what to do with indigenous peoples within the new nation's borders. For this reason, then, I mark the 1830s as the primary starting point for this book, with the turn toward a distinctly American scientific literature on race.

Ethnology, Gender, and American Science

From the start, ethnology was not a field on the fringes of the scientific establishment. On the contrary, its proponents held degrees or professorships from universities like Princeton, Harvard, the University of Pennsylvania, and Yale; maintained memberships in established and well-regarded organizations like the American Medical Association, the American Philosophical Society, and the Academy of Natural Sciences of Philadelphia; and gave lectures in such venues as the Smithsonian and the Royal Society of London. Nineteenth-century practitioners of ethnology should be termed *scientists*, not to lend credence to their long-discredited claims, but rather because their contemporaries regarded such work as legitimate and their qualifications and authority were rarely questioned, even by those who might have taken issue with their conclusions.

Though some racial theorists self-identified as ethnologists or spoke directly to their desire to see ethnology recognized as a distinct discipline, particularly in the early to mid-nineteenth century, racial science was a truly interdisciplinary endeavor with wide-ranging social and political concerns. A myriad of scientific disciplines were brought to bear on questions of race in the nineteenth century, including biology, comparative

anatomy, medicine, linguistics, Egyptology, anthropometrics, and, later in the century and into the twentieth, social hygiene, criminology, cultural anthropology, eugenics, and sexology as well. Moreover, many individual scientists bridged several of these fields in their professional interests or approaches. An abiding interest in human difference united these diverse scientific endeavors.

Ethnological writers were also engaged in a national conversation on race. To be sure, debates in the larger culture over slavery's efficacy informed many of the questions early ethnologists raised, but scientific critiques or endorsements of the institution did not break down neatly along regional lines, nor did such differences in political opinion preclude individual scientists from admiring—and citing—each other's conclusions on race more generally. That is, while ethnology developed as a field of scientific inquiry amid mounting political and cultural debate over slavery, individual ethnologists had complicated relationships with the institution. Numerous practitioners of ethnology themselves owned slaves and supported the institution personally and professionally; at the same time, some of the most adamant proslavery voices in ethnology were Northerners who did not themselves own slaves but defended racialized slavery in medico-scientific terms. Other proponents of ethnology in the North commented little on or even explicitly opposed the institution but seemed to welcome the national boost to their professional status when proslavery scientists or politicians cited their conclusions on racial difference. Still others criticized slavery as an abuse of a biologically inferior race, but they neither imagined nor promoted political equality between the races if slavery were to be abolished. Most white ethnological scientists across the political spectrum and regional divides insisted that biological distinctions between the races not only explained existing social and political hierarchies in America—and thus determined which races were worthy of citizenship—but also *necessitated* the social order.

While we do not typically think about scientists as mediating citizenship, that is precisely what the “scientists of race” attempted to do. As sociologists Michael Omi and Howard Winant remind us in their examination of racial formation in the second half of the twentieth century, “The state from its very inception has been concerned with the politics of race. For most of U.S. history, the state's main objective in its racial policy was repression and exclusion. Congress' first attempt to define American citizenship, the Naturalization Law of 1790, declared that only free ‘white’

immigrants could qualify. . . . Historically, a variety of previously racially undefined groups have required categorization within the prevailing racial order.”²⁴ The U.S. “scientists of race” provided such categorization and spoke directly to debates over who should be included—or, more often, excluded—from the category of citizen, bolstering their claims with biological assertions and scientific “evidence.” And in turn, at various moments in history when new groups sought to lay claim to citizenship rights, political leaders drew on scientific literature on race to justify their policies. Many scientists, for example, argued that African Americans were naturally servile, their bodies built for labor and their minds incapable of the reasoned thought needed for political involvement or leadership. Meanwhile, they touted the accomplishments of white men throughout history as the builders of great civilizations; the conquerors; the artists; and the writers of political doctrine, poetry, and prose. In the antebellum and Reconstruction eras, scientists often compared the “lower races” to women, particularly in intellectual capacity; their similarities to each other, in conjunction with the vast chasm that separated both from white men, illustrated that neither white women nor nonwhites were capable of political participation. Scientists in the late nineteenth and early twentieth centuries revised such characterizations in a far more ominous way, casting black men as a dangerous, menacing presence and white women as vulnerable outside the confines of domesticity, but that too functioned to underscore that neither had a place in American public life.

White men alone had a special biological mandate to full citizenship. Some scientists insisted they were merely categorizing the races and avoided commenting directly on politically charged issues like slavery or the franchise. However, their characterizations of the races as fundamentally different nonetheless had broad implications recognizable to their audience as well. In nineteenth- and early twentieth-century America, biology *was* destiny: different bodies inherently denoted different social roles—and by then, race and gender had replaced property ownership as the defining characteristics of U.S. citizenship. Historians Allison Sneider and Evelyn Nakano Glenn both detail the checkered history of citizenship in the nineteenth and early twentieth centuries, demonstrating that its definition and parameters were constantly being challenged—and ultimately reshaped—by women's suffragists as well as the larger socio-political developments of U.S. expansion, imperialism, immigration, and debates over free labor.²⁵ Faced with such challenges, racial scientists

continuously reinforced the de jure and de facto definition of citizenship as white and male. Consequently, the contested nature of citizenship—political as well as medical—is a theme that threads throughout this book and continues to resonate in U.S. politics in the twenty-first century. Michelle Alexander's groundbreaking book *The New Jim Crow: Mass Incarceration in the Age of Colorblindness* reminds us that the criminal justice system has replaced slavery and Jim Crow segregation as a means of controlling black men and maintaining their second-class citizenship status.²⁶ Meanwhile, the election, and subsequent two-term presidency, of Barack Obama saw many white voices rushing to declare America "postracial," others questioning his ability to lead utilizing rhetoric strikingly similar to ethnological scientists writing during Reconstruction, and still others (the so-called Birthers) challenging his U.S. citizenship entirely despite all evidence to the contrary, based in part on his racial heritage.

Likewise, racial science was essentially teleological. Rather than the strictly objective disciplines scientists often purported them to be, nineteenth- and early twentieth-century scientific endeavors on race took their cue from and reflected existing cultural ideologies and hierarchies. As feminist biologist Anne Fausto-Sterling reminds us, "'Science' is created by scientists." This seemingly simple statement underscores that *who* does science shapes the scientific knowledge that they produce. "Scientists are not neutral machines. They are social beings who have grown up in a particular society and have, through the process, acquired a particular framework or way of viewing the world," Fausto-Sterling notes. Further, "if science is made by non-neutral scientists, then it follows that science itself is not neutral. . . . Science is a construct created by a particular society which offers that society a particular view of the world. It changes and grows in response to societal changes, even ones which it may itself stimulate."²⁷ Similarly, as historians such as Laqueur have demonstrated in regards to sex and gender, scientists' expectations about race shaped what they saw in the body.

As a result, racial science was rife with claims that today seem contradictory and paradoxical, revealing the nature and extent of scientists' situational logic. For example, racial theorists often claimed that black men had longer arms proportional to white men, which they interpreted as a sign of the former's physical proximity to apes. But at the same time, they insisted that white men's greater amount of body hair was emblematic *not* of their own simian similarities but of their manliness. Beards could

be particularly evocative for racial scientists. New York physician and proslavery propagandist John Van Evrie, for instance, argued that white men had fuller beards than all other races. It is important to note, however, that the beard was a marker not just of racial difference for Van Evrie but also of manhood. Gender alone, though, as an analytic category, does not fully encompass or explain the implications of Van Evrie's claims, or many similar examples in ethnology tracts. For Van Evrie and many of his contemporaries, black men were irrevocably excluded from the category of citizen or political actor not simply because of their allegedly deviant gender performance, presumably a potentially adaptable behavior, but because their very biology excluded them from the male sex.

Putting Science to Work

From its beginnings in the early nineteenth century, racial science was an applied science. These scientists considered a range of political and social issues in their work, including slavery, the Reconstruction amendments, education, sexual variance, and violent crime. They examined Civil War soldiers for both health defects and racial characteristics, penned letters to the Freedmen's Bureau and editorials to newspapers, gave public speeches on America's "race problem," endorsed political parties or even specific candidates, testified in court in "sexual perversion" trials, served on vice commissions, treated (and mistreated) patients, and made recommendations on lynching. Racial science very much had a public life outside the scientific and medical establishment. As physician and prolific racial scientist G. Frank Lydston articulated in 1912, "The medical doctor is the best social doctor."²⁸

Accordingly, these scientists also directed their work toward ever-widening audiences. More and more over time, scientists articulated their conclusions about race and its sociopolitical implications in mass-produced pamphlets, university and medical school classrooms, mainstream periodicals, public lecture halls, and newspapers. And racial science indeed gradually *reached* larger segments of the public, in part because science and medicine had come of age, but also because the circulation of professional knowledge(s) in American culture had as well. The church and the increasingly professionalized sciences vied for cultural authority, while the nineteenth century also saw campaigns to bring both religion and education to the masses. With the growth of the publishing

industry that produced (and mass produced) books and periodicals at ever-cheaper prices, the proliferation of newspapers, tent revivals, lecture circuits, and traveling ministers, a far greater number of American citizens than ever before had access to information from a diversity of sources. The Chautauqua movement of the late nineteenth and early twentieth centuries, which brought an array of performers, religious speakers, scientists, and specialists of varying stripes to the public, embodied this trend toward mass education. Scientific theories about race reached a wide audience through such lecture circuits and print media; ethnology was a science *for* the people as much as it was *about* people.

These scientists often saw themselves as reformers, upholding the natural order and curing society of its ills. While their claims seem anything but progressive today—for example, some scientists' impassioned defenses of slavery or other scientists' advocacy of castration as a humane alternative to lynching at the turn of the century and into the 1920s—it is vitally important to examine racial scientists in their historical context, not to excuse them, but to better understand the culture that produced them and the role they played in American society. Politicians and cultural commentators cited racial science to buttress and legitimize their claims and policies, while scientific theories about race manifested themselves in popular visual culture, from political cartoons to campaign posters. In the nineteenth and early twentieth centuries, racial science was mainstream science.

It was never uncontested domain, however. Quite the contrary, even as scientific racism took shape over the course of the nineteenth century, so too did a vibrant counterdiscourse challenging it. Some white scientists challenged key tenets of racial science—or even, on occasion, the whole enterprise—as being built on tenuous logic or as inherently biased. More poignantly, throughout the long nineteenth century, African American intellectuals disputed scientific claims about black inferiority, argued for their place in the human family, and laid claim a proud history of great civilizations that their white counterparts tried to deny them—in other words, they offered their own theories about race and what it meant. Inheriting their intellectual mantle, Walter White, leader of the National Association for the Advancement of Colored People (NAACP), insisted that racial science was among the causes of racial violence and strove to dismantle scientific racism as an integral—but often overlooked—part of his antilynching work in the 1920s and 1930s. I end my story there, with White's antilynching work, the last spectacle lynching in the United States

in 1934, and the gradual decline of biological models of racial difference during this period as the influence of Franz Boas and cultural anthropology grew. The book, then, seeks to place racial science in a broader conversation in nineteenth- and early twentieth-century America, to unpack not just its genealogy but also its reception and social function. At turns embraced and contested, science was at the center of debates over the meanings of race, gender, and sexuality and the nature of power and privilege in the United States.

Race and Sex: Bodies under Threat

In the nineteenth and early twentieth centuries, the corporeal body—specifically, the sexed and sexualized body—was the primary site on which scientists examined the moral character and intellectual capacity of those Americans already outside the boundaries of the white, heteronormative nexus of social power and “respectability.”²⁹ The body held tremendous meaning for racial and sex differences, meanings that constantly shifted according to the specific context and historical moment that led a scientist to “read” it. This project, then, pays close attention to change over time, but it also reveals the persistence of certain ideas about race in American science in the nineteenth and early twentieth centuries—namely, that racial differences were biological and permanent and social hierarchies were a reflection of the natural order. In the antebellum period, scientists looked at bodies across racial lines to answer the “origins question” of whether the races constituted separate species, and some looked to naturalize slavery by interpreting black bodies specifically as biologically designed for servitude. In the late nineteenth century, evolutionary theory provided an ideological framework in which racial scientists compared and ranked bodies along a hierarchical ladder of human perfection in which white men always occupied the apex. And in the context of locating racial character and destiny on the physical body, the same scientists also began to examine the bodies of homosexuals for proof of their biological deviance.

Indeed, as the nineteenth century drew to a close, ethnological scientists were increasingly interested in sexual activity as it related to the nation's “race problem.” Concerned with a vast category they termed “sexual perversion,” America's most prominent racial scientists also became the country's first sexologists. Under this framework of “sexual perversion,” scientists were troubled by two manifestations in particular:

homosexuality among whites, which they saw as threatening the white race from within, and black male sexual aggression, which threatened the white race from outside. During the late nineteenth and early twentieth centuries, racial scientists offered solutions to the sexual threats they saw facing American society, threats that were also highly racialized.

Scientists' growing concern with homosexuality at the turn of the century was largely a product of their concern with the bodily integrity and sociopolitical status of the white race in the face of enormous social change in America. Similarly, more than any specific scientific discovery with regard to sexuality, scientists' reconceptualization of homosexuality as a medical pathology rather than a behavior, with supposedly correlated mental and corporeal imperfection, reflected their understandings of race and gender as biological entities. Scientists mapped these interrelated categories of difference—race, gender, sexuality—onto the body, as physical manifestations of existing social power dynamics and cultural hegemonies.

This book is informed by and in dialogue with a number of historical and scholarly literatures. Part of an emerging scholarship that examines the intersection of race, gender, and sexuality, it brings together histories that are often presented separately—including U.S. history, the history of race and ethnicity, African American history, cultural and intellectual history, the history of sexuality, women's and gender history, and the history of science and medicine. Though grounded in historical research and methodologies, my work also draws on and speaks to a number of interdisciplinary fields, including critical race studies, feminist science studies, gender studies, and queer studies, as well as the rich array of scholarship on the body within and across a variety of disciplines. It follows in the tradition of analyzing gender, race, and sexuality as cultural constructs—though one suspects that historians have only uncovered the tip of the proverbial iceberg when it comes to understanding the complex relationships among these constructed categories of difference.

The first two chapters provide a gendered analysis of scientific racism from the antebellum era through the turn of the century—a framework all too often missing from scholarship on American “sciences of race.” Seminal texts by George Fredrickson, William Stanton, Steven Jay Gould, John Haller Jr., and Winthrop Jordan have made undeniably valuable contributions to understanding racial thought in nineteenth-century America and continue to be widely cited today.³⁰ However, most of these

foundational works that address scientific racism were written prior to the rise of women's history as a subfield or, to borrow a phrase from historian Joan Scott, before gender became accepted as “a useful category of historical analysis.”³¹ The role of gender in ethnological understandings of racial difference, particularly during the mid-nineteenth century, is largely absent from these works. At the same time, this book is also indebted to more recent scholarship that has begun to move the historiography on scientific racism in exciting new directions. Nancy Stepan's work on gender, race, and science in England and Latin America, for example, has opened the door for a gendered analysis of scientific racism in the United States, which this book seeks to do.³² And Mia Bay's study of black thought about white racist science, and white people more generally, reminds us of the challenges of confronting hegemonic ideologies and just how high the stakes of ethnology were.³³

More than simply a corrective to earlier scholarship on scientific racism then, close attention to gender in medico-scientific thought on race challenges traditional understandings of the cultural politics of ethnology and demonstrates that the tone, focus, methodology, and intended audience of American racial science changed considerably over time, reflecting larger transformations in U.S. politics and society. As it did, scientists asked different questions about the sociopolitical implications of race and looked to different parts of the body for evidence of racial difference. Put simply, ethnology in the 1830s and 1840s tended to focus on more gender-neutral features like skulls, hair, and skin (itself all the more striking considering European scientists' fascination with Africans' sexualized bodies and behaviors earlier in the century and prior), while in the years surrounding the Civil War, when scientists took up the profoundly gendered issues of slavery and citizenship, they also featured secondary sex characteristics such as beards and breasts in their bodily analyses. And after Reconstruction, American racial scientists even more overtly engaged not just with gender but also with sexuality and sexualized bodies. Amid frequent discussions of miscegenation and “race suicide,” as well as the rise of eugenics—a field that quite literally embodied the nexus of the “sciences of race” and the “sciences of sex”—they added reproductive organs to the list of physical characteristics they investigated for evidence of racial difference. That is, sexual features were *socially* relevant to ethnological scientists' readings of race at the turn of the century in ways they were not earlier in the nineteenth century. While which body parts most

received racial scientists' attention changed over time, to some extent so too did what each part signified. One thing changed very little, however: no matter where in the body white racial scientists looked and when, they interpreted what they saw through the lens of pervasive difference rather than similarity.

These developments in ethnology were profoundly interconnected in ways not yet explored by historians of nineteenth-century America. While George Fredrickson has highlighted the increasingly proslavery stance of many ethnological theorists beginning in the 1850s, historians have not examined the impact of this development on gender in scientific considerations of race, for example.³⁴ Indeed, historical accounts of racial science before the late nineteenth century have paid little attention to gender. Scholarship on racial thought at the turn of the century, on the other hand, has examined the construction of black men as sexual predators during that period and analyzed the centrality of sex and gender in evolutionary theory; however, this scholarship largely attributes scientists' attention to gender to the rise of social Darwinism rather than an extension of earlier developments. The first two chapters, in contrast, reveal that gender *was always* integral to scientific understanding of race from the start, but the function it served changed over time with shifts in the nation's cultural and political landscape. Consequently, they detail a series of "call-and-response" moments in which scientists reacted to and offered their expertise on these shifts while underscoring ethnology as an interracial site of contestation and race making.

Chapter 1 traces the rise of biological paradigms of racial difference and the development of ethnology as a field of scientific inquiry. It argues that early ethnology was entirely male dominated; it focused on men, presumed a male audience, and was produced by male scientists. For example, the question of racial origins—the dominant concern in ethnology prior to the Civil War—was framed in terms of male lines of descent. Scientists debated whether the black and white races alike descended from Adam (rather than Adam and Eve) or if the black race could be traced to Ham, the cursed son of Noah—a racial theory that remains alive and well among modern white supremacist organizations like the Aryan Nation. At the same time, black intellectuals directly challenged the tenets of scientific racism, particularly the theory of polygenesis that placed African Americans outside of the human family, and created their own models

of race—though these too were focused on men. When racial scientists did discuss women, it was usually in the context of their considerations of slavery, which often pointed to slave women's reproduction as indicative of the system's overall health and viability. Ethnology's adherents never presented a united front on America's "peculiar institution," however, and prior to the 1850s and 1860s, when the issue heated up in U.S. political culture in response to the increasingly powerful abolitionist movement and debates about the expansion of slavery into the Western territories, it was common for racial scientists to ignore the issue entirely. As this chapter reveals, gender was central to ethnological arguments both for and against the institution. As with other issues in antebellum ethnology, scientists focused more often on men, but "man" itself was a normative category that was subject to little scrutiny. With few exceptions, it was not until the Civil War that scientists began to explicitly analyze manhood—its physical parameters and political import—vis-à-vis questions of race.

Chapter 2 describes this transitional moment in American racial science, when scientists became deeply invested in defining what manhood meant and to whom it applied.³⁵ During the Civil War, military scientists conducted large-scale anthropometric studies of union troops, quite literally measuring manhood along racial lines, and attempted to ascertain what, and who, constituted the ideal soldier in both body and character. Their interest extended beyond the military context, however, and they also sought to qualitatively and quantitatively define the "average man" and, by extension, the best man. During Reconstruction, ethnological scientists turned their attention from the question of black men's fitness for military service to their fitness for full citizenship, a political category attached to men alone. Many argued against black male suffrage by insisting they were not really "men" in body or behavior, and scientists often compared black men and women as having similarly limited intellectual capacities to illustrate that neither deserved the vote. With Emancipation and black and women's suffrage posing a challenge to the traditional American definition of citizenship, scientists rallied to reassert the citizen body as inherently both white and male. The economic and political competition that the freed black population had posed during Reconstruction was met with violent backlash in the period that followed, as the South was "redeemed" for white supremacy in the late nineteenth century. More generally, white scientists were concerned by reproductive competition

and threat as well, that the nation's balance of power could come down to a contest of numbers and that miscegenation represented a menace to American civilization—a sharp contrast to antebellum racial theories that minimized the realities of sexual exploitation under slavery by writing off mixed race people themselves as inconsequential, sterile, and destined to die off. In this context, with the integration of evolutionary theory and social Darwinism into American racial science, ethnological critiques of black men's capacity for citizenship became more sexualized in the late nineteenth and early twentieth centuries. The American obsession with race and reproduction, in popular and political discourse as well as scientific thought, during this period intersected with the nation's imperial project, debates over immigration, anxieties about the free black population, and the eugenics movement.³⁶ This chapter situates that obsession within a larger chronology of scientific thought in which scientists continually adapted their use of gender and sex in conceptualizing race to new political and cultural circumstances and looked to different parts of the body for racial truths with each shift.

At a time when many Americans believed that reproduction and scientifically strategic breeding would determine the fate of civilization and white supremacy, turn-of-the-century scientists were quite preoccupied with sex. They subsumed under a vast umbrella category a polyglot of sexual acts, crimes, and social trends—real and imagined—they perceived as threatening the natural order. And indeed, American society *was* changing in the wake of Emancipation and the advancements of Reconstruction; the waves of immigrants flooding U.S. cities and the workforce; and women's rights advocates pushing for educational opportunities, the franchise, and the ability to control their own reproduction. For their part, scientists, representing a range of subfields, medical specialties, and professional disciplines, attempted to combat this gender and racial disorder—which challenged their own position at the top of a white patriarchal hierarchy—by controlling sex. Chapters 3 and 4, then, examine the ways in which scientists sought to maintain their authority and cultural relevance by speaking to the nation's intersecting anxieties around race and sex and promoting “practical,” medically based solutions for the nation's social ills.

Scholars are just beginning to explore the intersections of race and sexuality—and queer sexualities in particular—in U.S. history. Lisa

Duggan, Siobhan Somerville, Kevin Mumford, Julian Carter, Sharon Patricia Holland, and Nancy Ordoover have raised important questions about these intersections in American culture, revealing exciting new insights by applying queer theory to studies of race.³⁷ Duggan, Somerville, Carter, and Ordoover also underscore the racial politics driving scientific studies of sex and the pathological models of same-sex sexuality that emerged at the turn of the century, a topic I take up at length in chapter 3, which extends and deepens such analyses by demonstrating that racial science and sexology were not separate fields in America. They not only shared similar methodologies and concerns, as previous scholars have pointed out, but they literally shared the same key scientists. Moreover, although turn-of-the-century scientists disagreed over the distinctions or overlap between homosexual acts, gender deviance, and physical hermaphroditism, scientific assessments of the causes of “sexual perversion” frequently broke down along racial lines. “Sexual perversion” among nonwhites and the lower classes was characterized as vice, indicative of the physical and moral degeneracy of the group. In contrast, scientists usually read cases of “sexual perversion” or inversion in middle- and upper-class whites as symptomatic of individual pathology or disease, which many attributed to “overcivilization.” While African Americans, immigrants, and the lower class threatened moral corruption through the spread of their sexual “vice,” white homosexuals from respectable families threatened “race suicide.”

Meanwhile, chapter 4 argues that emasculation was central to both scientific and popular discourse on lynching as well as its practice. The waves of lynchings and the demonization of black men as rapists during the late nineteenth and early twentieth centuries are well known. But less well known are scientific responses to lynching. For more than three decades, numerous American scientists recommended surgical castration as an alternative to lynch violence. Far from marginal, these prominent scientists had a profound impact on American society outside the medical establishment. They lent scientific support for the “black beast rapist” trope but positioned themselves as progressive reformers offering a medical solution to the problem, spurned on in part by increasingly negative publicity on lynching itself. In so doing, they sought to place the authority over America's “race problem” in the hands of scientists rather than mobs. Meanwhile, lynch mobs often included castration as a crucial part of—rather than a substitute for—the lynching spectacle. Rather than an

anomaly, the "castration remedy" represented a culmination of scientists' intersecting concerns with race and sex, as well as their role as "social doctors" fixing America's ills.

Not everyone thought science offered the cure to America's issues around race, however. In the 1920s and 1930s, NAACP antilynching activist Walter White identified racial scientists as part of the problem, not the solution. Sharing an intellectual genealogy with nineteenth-century black thinkers who challenged the core tenets of ethnology, he pointed to the impact of what was by then a century of scientific racism on the way living, breathing human beings were treated in American society. As demonstrated in chapter 5, White made debunking biological theories of black inferiority a key component of his antilynching work with the NAACP and sought strategic alliances with sympathetic scientists who could help him turn science against itself. That he was able to build such alliances with prominent *white* scientists in particular was a testament to his tenacity, to be sure, but also signaled a crucial moment in American history in which the logic of racial science was increasingly under attack. Throughout his work, White critiqued the black rapist trope that so often served to justify lynching, but he shrewdly robbed this sexualized rhetoric of the legitimacy it had long received from science by attributing it only to ignorant mobs rather than physicians' pens. While he was not the first black thinker to suggest that scientific racism precipitated racial violence, White had something his predecessors did not: powerful white friends, political influence through his leadership in the NAACP, and a wide audience that crossed the color line.

Above all, this book is a story about how categories of human difference are created, maintained, and contested and the role of science in that process. Throughout much of the nineteenth century and into the twentieth, American scientists were engaged in a complex examination of race, sex, gender, and sexuality. These categories were perpetually in flux, and the body was the site on which they were built. Ultimately, they were not just categories of difference, however; they were, and often continue to be, categories of sociopolitical exclusion. Accordingly, several interrelated scholarly and historical concerns have driven this work, including the function of science in American society; the development of racial science as a field and its authority over sociopolitical issues and debates; the racialization of homosexuality and the concurrent rise of sexology; challenges

to scientific claims about racial inferiority; the social construction and intersection of race, gender, and sexuality; and the body as a cultural entity and site of contestation. But while scholars generally talk about these categories as social constructs, they continue to have undeniable cultural and material salience in America.